

Multi-Agent Marketing Strategy Platform

Using OpenAI Agents SDK

SUMMER SCHOOL '26

OPENAI AGENTS SDK - CAPSTONE PROJECT

Technology: Python | OpenAI Agents SDK | Gradio

Business Input

Provide goals and raw data

AI Agents

Autonomous agents analyze inputs

Marketing Strategy

Generate targeted campaign plans

Human Approval

Review and refine recommended plans

Analytics

Measure performance and feedback

This platform transforms raw business inputs into actionable marketing strategies through coordinated AI agents, with human oversight at every critical decision point.

Problem Analysis – Business Context

Modern marketing demands execution across multiple disciplines – market research, competitor analysis, campaign planning, content strategy, optimization, and performance analytics. Managing these tasks separately requires significant time, coordination, and specialized expertise. The AI Marketing Strategy Manager solves this by assigning each task to a specialized AI agent and combining their outputs into one unified marketing strategy.

Traditional Process Multiple Manual Tasks → Separate Analysis → Manual Strategy	AI-Powered System Business Input → Specialized AI Agents → Unified Strategy
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Marketing Activities Covered

- Market research
- Competitor analysis
- Campaign planning
- Content strategy
- Marketing optimization
- Campaign performance analysis

The Core Challenge

Without an integrated system, marketing teams must juggle multiple tools and manual processes – leading to fragmented strategies, inconsistent outputs, and slower decision-making. The AI Marketing Strategy Manager consolidates all of this into a single, coordinated workflow powered by specialized AI agents.

Stakeholders, Problem & Objectives

Stakeholders

- Business owners
- Marketing managers
- Marketing teams
- Campaign managers
- Content teams
- Decision makers

Problem Statement

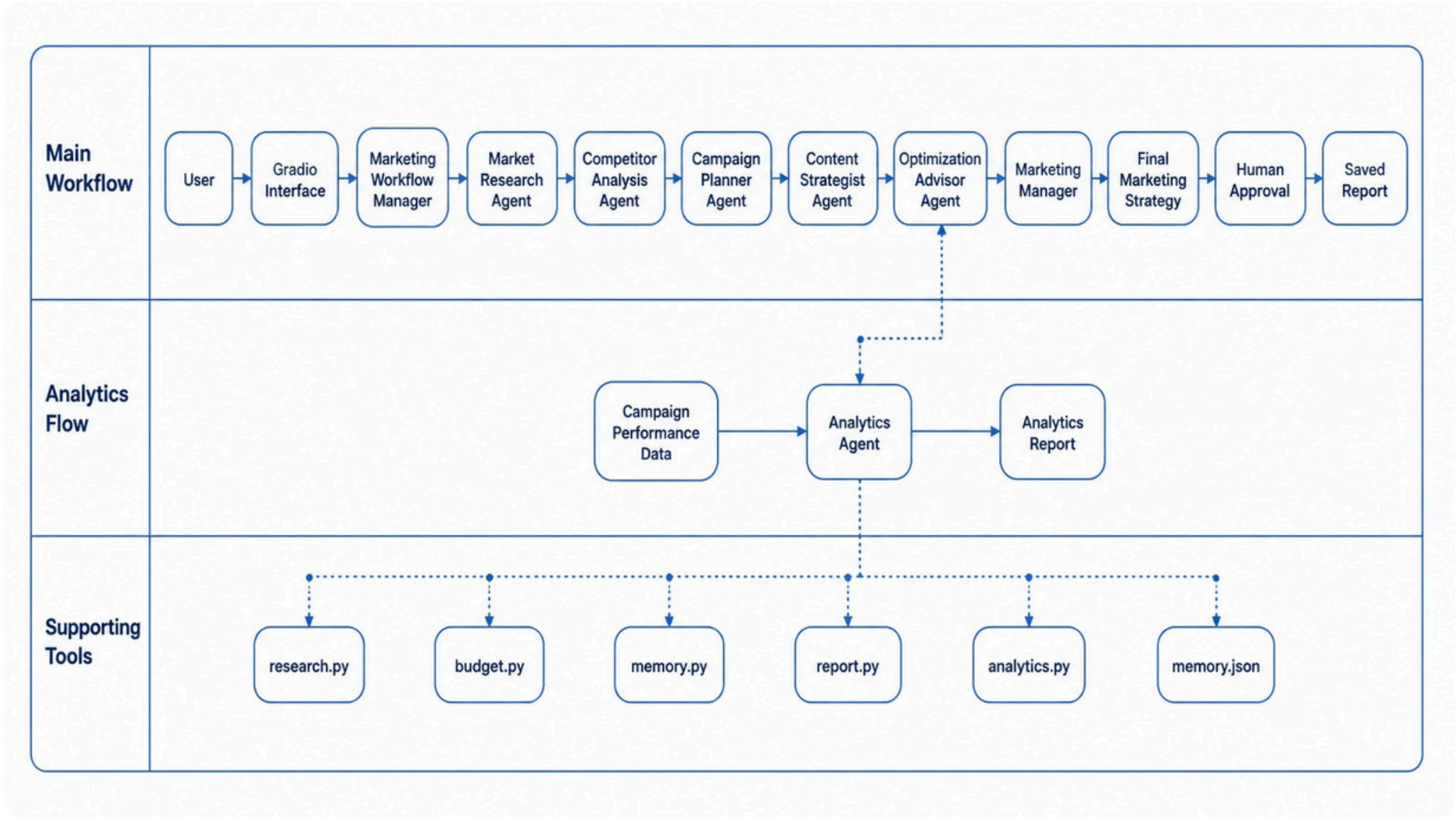
Develop a multi-agent AI marketing platform capable of researching markets and competitors, planning campaigns, creating content strategies, recommending improvements, and analyzing campaign performance — all within a single coordinated system.

Project Objectives

Automate Major marketing strategy tasks	Specialize Use dedicated AI agents per function
Coordinate Combine agent outputs seamlessly	Structure Generate organized, complete results
Approve Maintain human oversight	Analyze Track actual campaign performance

Multi-Agent Architecture

The platform follows a layered architecture where the Gradio interface captures business input, the Marketing Workflow orchestrates specialized agents in sequence, and the Marketing Manager consolidates outputs into a final strategy. A separate Analytics Agent handles campaign performance data independently. Supporting tools – research.py, budget.py, memory.py, report.py, analytics.py, and memory.json – provide persistent context and structured output generation throughout the workflow.



The architecture separates the main strategy generation pipeline from the analytics pipeline, allowing each to operate independently while sharing context through the memory system.

Specialized AI Agents

Each agent is designed with a focused responsibility, ensuring depth and quality in every area of the marketing strategy. The Marketing Manager then synthesizes all specialist outputs into a cohesive final strategy.



Market Research Agent

Analyzes target market, customer needs, trends, and opportunities.



Competitor Analysis Agent

Analyzes competitors, strengths, weaknesses, and market positioning.



Campaign Planner Agent

Creates campaign objectives, channels, budget allocation, timeline, and KPIs.



Content Strategist Agent

Develops marketing content ideas, messages, and channel strategies.



Optimization Advisor Agent

Reviews the proposed marketing strategy and recommends improvements.



Analytics Agent

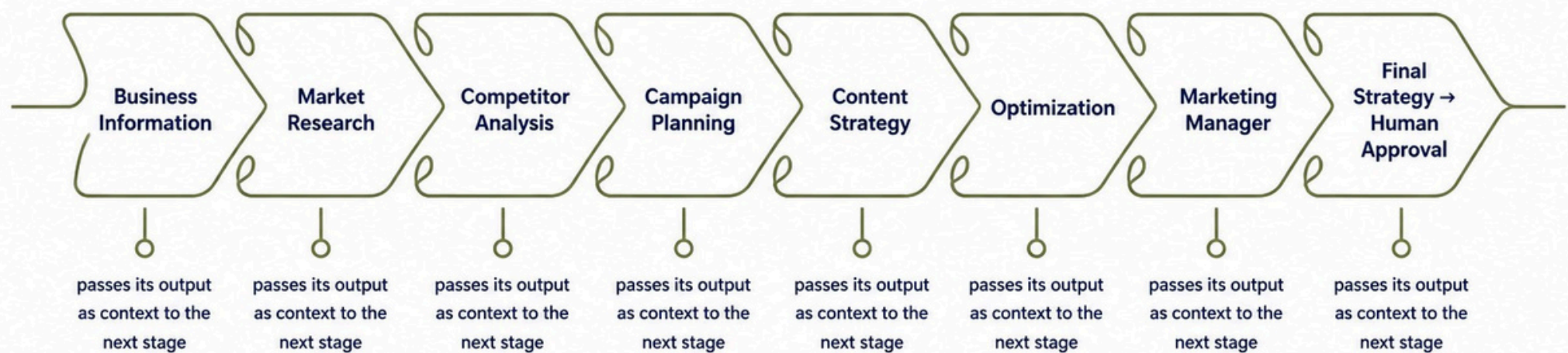
Analyzes actual campaign performance data and generates analytics reports.



Marketing Manager: Combines all specialist agent outputs into the final professional marketing strategy presented to the user.

Agent Interaction and Workflow Flow

Each specialized agent performs a specific marketing task in sequence. The workflow coordinates the agents and passes relevant outputs from one stage as context to subsequent stages. This ensures continuity and consistency throughout the marketing strategy generation process. The Marketing Manager combines the specialist outputs into the final marketing strategy, which is then presented to the user for human approval.



Context Passing

Each agent receives the accumulated context from all previous stages, enabling informed and consistent recommendations throughout the strategy generation process.

Human Approval Gate

The final strategy is presented to the user for review and approval before being saved. This ensures business owners and marketing managers retain full control over the output.

Tools and Technology Integration

The platform is built on a focused technology stack designed for AI agent orchestration and interactive user experience. Python serves as the core language, OpenAI Agents SDK handles agent creation and coordination, and Gradio provides the interactive web interface. Custom tool modules support specific marketing operations throughout the workflow.

Tool / Technology	Purpose
research.py	Market research and research-related operations
budget.py	Marketing budget calculations and validation
memory.py	Store and retrieve business/context information
report.py	Report generation and saving
analytics.py	Campaign performance calculations and analytics
memory.json	Persistent business/context storage
Python	Core programming language
OpenAI Agents SDK	AI agent creation and orchestration
Gradio	Interactive web interface

Python

Core language powering all agent logic, tool integrations, and workflow orchestration

OpenAI Agents SDK

Enables creation, coordination, and handoff between specialized AI marketing agents

Gradio

Provides an interactive web interface for users to input business data and review strategies

Memory and Structured Outputs

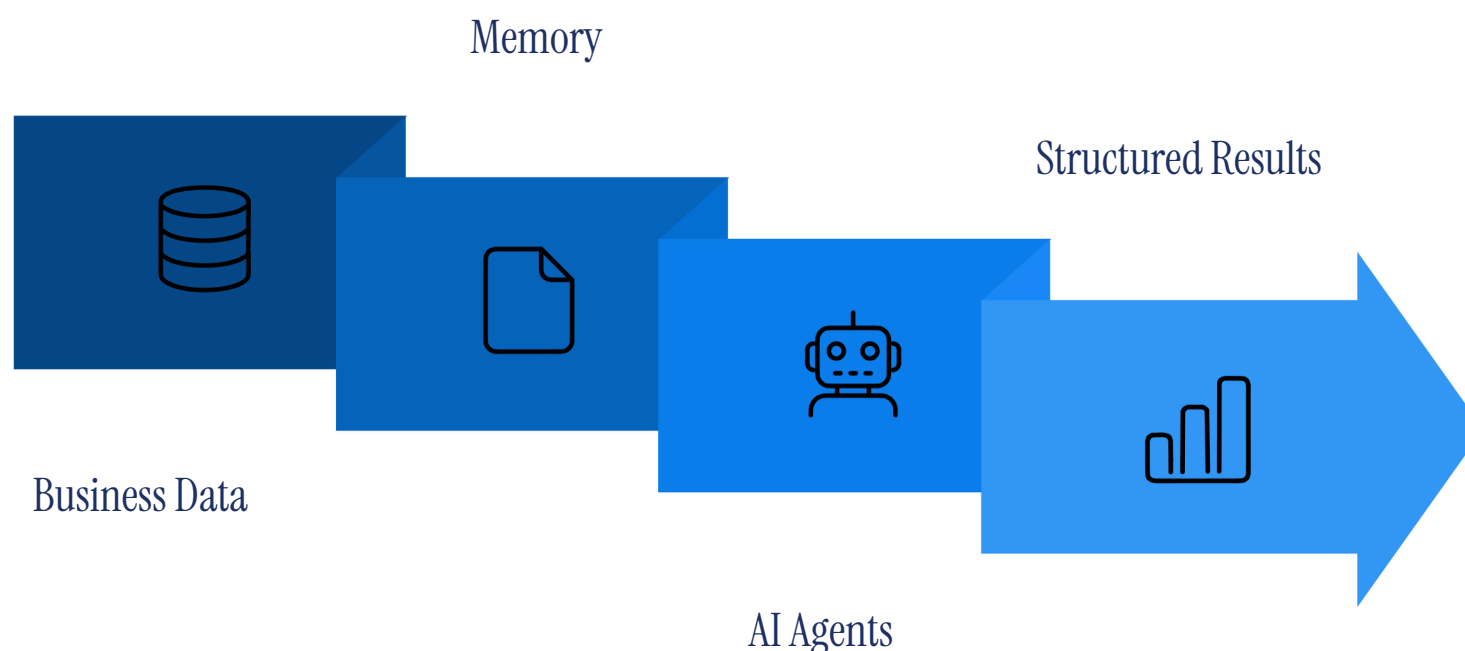
Memory & Context Management

The project uses memory.py and memory.json to manage business information and context throughout the workflow. Relevant data is stored and reused across agent stages, ensuring continuity and consistency in the final strategy.

Structured Output Sections

Agent results are organized into clearly defined marketing sections:

- Market Research
- Competitor Analysis
- Campaign Plan
- Content Strategy
- Optimization Recommendations
- Final Marketing Strategy
- Analytics Report



This flow illustrates how business data flows through the memory system and AI agents to produce organized, structured marketing outputs ready for review and approval.

Human Approval and Campaign Analytics

Human Approval Workflow

AI-generated strategies are never deployed automatically. The user reviews the complete marketing strategy before granting approval, maintaining full control over the output.

1. AI Agents generate the Final Marketing Strategy
2. User reviews the complete strategy
3. User approves the strategy through the Gradio interface
4. Approved strategy is saved to the outputs folder

Campaign Analytics

After campaign launch, users enter performance data and the Analytics Agent calculates key marketing metrics automatically.

Input Metrics:

- Impressions
- Clicks
- Conversions
- Advertising Cost
- Revenue Generated

Calculated Metrics:

- CTR (Click-Through Rate)
- Conversion Rate
- Cost Per Conversion
- ROAS (Return on Ad Spend)

The Analytics Agent also provides performance analysis and marketing recommendations based on the calculated results.

Project Summary

The AI Marketing Strategy Manager is a multi-agent platform that automates and coordinates key marketing strategy tasks — from market research and competitor analysis to campaign planning, content strategy, optimization, and performance analytics. Built with Python, OpenAI Agents SDK, and Gradio, it delivers structured, human-approved marketing strategies through specialized AI agents working in sequence.

Technology Stack

Python | OpenAI Agents SDK | Gradio

7 Specialized Agents

Market Research, Competitor Analysis, Campaign Planner, Content Strategist, Optimization Advisor, Analytics, Marketing Manager

Human-in-the-Loop

Full user review and approval before any strategy is saved or deployed

End-to-End Workflow

From business input to structured strategy to campaign analytics — all in one platform

The AI Marketing Strategy Manager demonstrates how specialized AI agents, coordinated through the OpenAI Agents SDK, can transform fragmented marketing workflows into a unified, intelligent strategy generation system — with human oversight at every critical step.